

# Virtual Assistant Track Machine Learning Model

## Analysis Report

**Prepared by:** Rofiat Muhammad

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### Executive Summary

This report analyzes the performance of **196 participants** in the HNG i13 Virtual Assistant track across **five assessment stages** (Stages 1, 2, 3, 4, and 7). The analysis combines descriptive statistics, stage-level performance review, attrition analysis, and a machine learning model to predict participant success.

Key findings show that while **68.37%** of participants passed at least one completed stage, only **41.53%** reached the final stage. The overall **average grade is 2.68/10**, indicating a highly selective and demanding pipeline. Only **39 participants (19.9%)** qualified as high performers.

A Random Forest model was trained to predict whether participants would pass all stages they attempted. The model achieved **95% accuracy** and a **ROC-AUC score of 0.98**, demonstrating strong predictive reliability. Early-stage performance and consistency emerged as the strongest indicators of long-term success.

### Data Overview

The dataset was consolidated from five separate Excel score sheets, one for each stage. After merging and cleaning:

- **Original merged records:** 309
- **Unique participants:** 196
- **Stages tracked:** 5
- **Total features:** 28 (after feature engineering)

Duplicate records caused by name and Slack ID variations were resolved by aggregating best performance values per participant. Missing stage grades were filled with zero strictly for

analytical consistency. In the dataset, I only worked with Stage 1,2,3,4, and 7 because there was no individual stage task for stage 5 and 6.

## Feature Engineering and Metrics

Key metrics were derived to support analysis and modeling:

- **average\_grade:** Mean grade across completed stages
- **grade\_improvement:** Difference between Stage 7 and Stage 1
- **completion\_rate:** Percentage of stages completed
- **grade\_std:** Performance consistency measure
- **total\_passes / total\_fails:** Stage outcomes count
- **performance\_category:** Needs Improvement, Satisfactory, Excellent

Participants were segmented using data-driven thresholds, enabling performance comparison and targeted analysis.

## Overall Performance Analysis

Performance across the cohort is strongly skewed toward low scores:

- **Average grade:** 2.68
- **Median grade:** 1.65
- **Participants scoring  $\leq 2.5$ :** 60%
- **High performers:** 39 participants (19.9%)

This distribution confirms that the evaluation process filters aggressively and rewards consistent, sustained performance rather than isolated high scores.

## Stage Progression and Attrition

Attrition occurs at every stage, with two major drop-off points:

- **Stage 1 → Stage 2:** 33.5% dropout
- **Stage 4 → Stage 7:** 53.2% dropout

Although Stage 4 records the highest average grade, a significant portion of participants fail to transition to the final stage. This suggests that late-stage difficulty and participant fatigue may be contributing factors.

Consistency proved more important than peak performance. Participants who maintained stable grades across stages were far more likely to complete the pipeline.

## Machine Learning Model

A **Random Forest Classifier** was developed to predict whether participants would pass all completed stages.

### Model configuration:

- Trees: 100
- Max depth: 5
- Train-test split: 80/20 (stratified)

### Performance results:

- **Accuracy:** 95%
- **ROC-AUC:** 0.98
- **Recall (Pass):** 100%
- **Recall (Fail):** 85%

The model makes no false-negative predictions for successful participants, meaning high-potential candidates are not missed.

## Key Predictors of Success

Feature importance analysis reveals the strongest drivers of success:

1. **average\_grade** – 22.7%
2. **Stage 4 Grade** – 17.9%
3. **grade\_improvement** – 16.0%
4. **Stage 1 Grade** – 12.5%

Early performance sets the trajectory, but improvement over time plays a critical role. Participants who show learning progression outperform those with strong starts but declining trends.

## Prediction Segmentation

Participants were grouped into three predictive outcomes:

- **Will Finish HNG:** High probability of success
- **Might Finish HNG:** Moderate risk
- **Will Go Home:** High risk of attrition

Approximately **25% of participants** fall into the high-risk category and can be flagged early for intervention. A smaller subset of “hidden gems” displays strong improvement despite low average grades, highlighting candidates worth retaining.

## Key Insights and Findings from Dashboard

### Overall Participation & Performance

- 196 participants entered the pipeline.
- 68.37% passed at least one stage, showing strong initial engagement.
- The average grade across all stages is 2.68 (out of 10), indicating the pipeline is challenging and selective.
- Only 39 participants (19.9%) are classified as “High Performing Interns”, these are your top talent.

Insight: The program successfully identifies high performers early, but the low average grade suggests many struggles with the difficulty level.

### Stage-by-Stage Attrition

- Stage 1: 178 participants (100% of those who started)
- Stage 2: Drops to 38 participants (only 21.3% of Stage 1)
- Stage 3: 80 participants (45% of Stage 2)
- Stage 4: 77 participants
- Stage 7 (Final): Only 35 participants remain; just 19.7% of the original cohort.

Insight: There's a massive drop-off after Stage 1. Stage 2 is the biggest bottleneck; only 21% make it through. This suggests Stage 2 may be too difficult or poorly designed.

### **Performance Distribution**

The performance categories show:

- 59.69% → Needs Improvement (Low performers)
- 20.41% → Satisfactory (Average performers)
- 19.9% → Excellent (Top performers)

Insight: Most participants need support. Focus resources on the “Needs Improvement” group to boost overall success.

### **Grade Progression Across Stages**

- Stage 1 Avg: 6.33
- Stage 2 Avg: Peaks at 7.03 (strongest performance)
- Stage 3 Avg: Drops sharply to 6.21 (biggest dip)
- Stage 4 Avg: Recovers to 6.52
- Stage 7 Avg: Slightly up to 6.56

Insight: Participants perform best in Stage 2, then struggle in Stage 3. This could indicate Stage 3 is the most challenging or least aligned with participant skills.

### **Machine Learning Model Predictions & Risk Alerts**

- 55 participants are flagged as “At Risk” (predicted to not finish).
- Average success probability for the entire cohort is 0.68 (68% chance of success).
- Final Success Rate (Stage 7 Pass) is 10.71%; very low, highlighting how hard it is to complete the full pipeline.
- The model predicts 62.76% will “Finish HNG”, 25.5% “Might Finish”, and 11.73% “Will Go Home”.

Insight: The ML model is a powerful tool for early intervention. Use the “Risk Alert Table” to identify and support the 55 at-risk participants before they drop out.

### **Top Performers & Pass Rates by Stage**

Top 10 Interns (by average grade) range from 6.8 to 8.3 these are your star performers. Pass Rates per Stage:

- Stage 1: 91%
- Stage 2: 82%
- Stage 3: 71%
- Stage 4: 66%
- Stage 7: 60%

Insight: Even the strongest candidates face challenges — Stage 7 has a 60% pass rate, meaning 40% of even good performers fail the final stage.

## Recommendations

1. **Early Intervention:**

Flag participants with low Stage 1 and Stage 2 grades for immediate mentorship.

2. **Stage 2 Review:**

Address the early pipeline bottleneck through improved guidance or transitional support.

3. **Stage 4 Focus:**

Reinforce Stage 4 as a critical checkpoint with targeted preparation and coaching.

4. **Predictive Monitoring:**

Embed ML predictions into a live Power BI dashboard for real-time risk tracking.

5. **Support Consistency:**

Reward stable performers, not just high scorers, to encourage sustained effort.

## Conclusion

This analysis converts raw evaluation data into actionable insight. By combining performance metrics, stage-level analysis, and machine learning predictions, the HNG program can identify talent earlier, reduce avoidable attrition, and focus resources where they have the greatest impact.

The framework established here is scalable, repeatable, and suitable for future cohorts, supporting a more efficient and data-informed selection process.